

RESEARCH PROPOSAL

Intelligent Tutor Based on Multi-Agent Architecture and Large Language Models for Adaptive Learning of Ancash Quechua

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Main Research Question

How can a multi-agent architecture integrated with Large Language Models (LLMs) enable the development of an adaptive intelligent tutor to optimize the learning process of Ancash Quechua in university students?

2. Method-Fit Matrix (E.D.F.C.V.)

CRITERION (weight)	Pure Ex- perimental	Pure Case Study	DESIGN SCIENCE (DSR + Eval)	Why this score
E · Epistemological fit (30)	2 / 5	2 / 5	5 / 5	The core goal is to build and evaluate an intelli- gent artifact. DSR vali- dates IT/AI innovation.
D · Data availability (20)	3 / 5	4 / 5	4 / 5	Regional corpora and le- xicographical data exist. Access to students for adaptive evaluation is feasible.
F · Feasibility (15)	4 / 5	5 / 5	3 / 5	Orchestrating LLM/RAG agents with Ancash Quechua data requires engineering ti- me. Honest score.
C · Contribution type (20)	3 / 5	2 / 5	5 / 5	Delivers a functional ar- tifact and a multi-agent architecture adapted to a low-resource language.
V · Venue fit (15)	3 / 5	2 / 5	5 / 5	AI and education venues (<i>Computers & Educa- tion, IEEE TLT</i>) priori- tize DSR with empirical evaluation.
GATE ($E \leq 2$ eli- minates)	ELIMINATE	ELIMINATE	SURVIVES	<i>The gate automati- cally eliminates pure options due to lack of epistemological fit with AI/software.</i>
Weighted total	14.0 / 25 raw	12.5 / 25 raw	★ 22.0 / 25 raw	<i>Both experimental and case study are gated out ($E \leq 2$). Design Science Research wins.</i>